

Optimal simplex products for projection-body volume

Exact extremizers, periodicity, and a sharp isolation gap

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Abstract

We determine the maximum of normalized projection-body volume over all affine images of Cartesian products of simplices in every dimension. The unique optimal exponential block size is thirteen. In every total dimension, only two balanced products need be compared. From dimension one hundred onward, the maximizing product is given by an exact period-thirteen rule, using thirteen-dimensional factors and at most eight fourteen-dimensional or four twelve-dimensional factors. We prove uniqueness of the maximizing dimension multiset and a sharp, dimension-independent multiplicative gap to every other product. The optimum satisfies an exact recurrence and has an explicit periodic asymptotic expansion relative to the simplex value, with optimal exponential base approximately 1.03372825553. All required finite inequalities are certified by integer arithmetic, and an independent exact dynamic program checks the formulas through dimension three hundred. The results refine a recent product construction; they make no claim about the unrestricted maximum over all convex bodies. The geometric reduction and every optimization argument are proved in full.

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1 Introduction and scope

For an n -dimensional convex body K , write

$$R_n(K) = \frac{|\Pi K|}{|K|^{n-1}}, \quad s_d = \frac{(d+1)d^d}{d!}.$$

The projection body is defined by $h_{\Pi K}(u) = |\text{proj}_{u^\perp} K|_{n-1}$ for unit u . We consider the class of invertible affine images of Cartesian products of nondegenerate simplices. The factor dimensions can vary, and their number is unrestricted. The results in this section optimize over that class; they do not determine the maximum over all convex bodies.

The source manuscript [1] proves the two identities

$$R_d(T_d) = s_d, \quad R_{r+s}(A \times B) = R_r(A)R_s(B), \quad (1.1)$$

where T_d is any nondegenerate d -simplex. Section 3 rederives them for completeness. The source uses repeated ten-dimensional blocks to obtain an exponential improvement over a single simplex. We optimize that construction exactly, identify its unique optimal exponential block size, and give a sharp isolation gap between optimal and nonoptimal products. The product counterexample itself is not new: the source explicitly cites the earlier counterexamples of Feng, Hu, Liu, and Xu [3] to the unrestricted simplex-maximum conjecture of Brannen [2].

Let

$$P_n = \max \left\{ \prod_{i=1}^k s_{d_i} : k \geq 1, d_i \geq 1, \sum_{i=1}^k d_i = n \right\}. \quad (1.2)$$

In dimension one the natural convention is $\Pi I = [-1, 1]$ and $s_1 = 2$. The facet proof of (1.1) extends to an interval factor by assigning volume one to a point. Alternatively, interval factors can be omitted throughout: whenever $n \geq 2$, merging a dimension-one factor with another factor strictly increases the objective, as follows from the ratio formula below. Thus both conventions give the same P_n for $n \geq 2$.

2 Exact optimization

Define

$$\rho = s_{13}^{1/13}, \quad L = \log \rho, \quad \alpha = 12L - \log s_{12}, \quad \beta = 14L - \log s_{14}. \quad (2.1)$$

The constants α and β are positive. Numerically,

$$\rho \simeq 2.80996473256, \quad \alpha \simeq 0.001448531660, \quad \beta \simeq 0.000775423091.$$

The decimals are descriptive; all inequalities used in the proof have exact rational certificates.

Theorem 2.1 (Exact eventual optimizer). *Write $n \equiv r \pmod{13}$, with $0 \leq r \leq 12$. For every $n \geq 100$,*

$$P_n = \begin{cases} s_{13}^{(n-14r)/13} s_{14}^r, & 0 \leq r \leq 8, \\ s_{12}^{13-r} s_{13}^{(n-12(13-r))/13}, & 9 \leq r \leq 12. \end{cases} \quad (2.2)$$

Every maximizing factorization has exactly the indicated dimensions, up to ordering. Consequently the maximizing bodies in the stated class are uniquely specified up to invertible affine maps.

The first formula is valid as soon as $n \geq 14r$; the second is valid as soon as $n \geq 12(13 - r)$. The uniform threshold 100 is the smallest threshold for which these residue formulas hold in every dimension. Moreover,

$$\sup_{d \geq 1} s_d^{1/d} = s_{13}^{1/13} = \rho, \quad (2.3)$$

and equality in (2.3) occurs only at $d = 13$.

For all dimensions, not merely those beyond the periodic threshold, a two-candidate formula suffices. Given $1 \leq k \leq n$, put

$$a = \lfloor n/k \rfloor, \quad b = n - ka, \quad B(n, k) = s_a^{k-b} s_{a+1}^b.$$

Theorem 2.2 (Two candidates in every dimension). *For every positive integer n ,*

$$P_n = \max\{B(n, k) : k \in \{\max(1, \lfloor n/13 \rfloor), \lceil n/13 \rceil\}\}.$$

Repeated candidates are counted once. In particular, computing the optimal value requires comparing only two explicit rational numbers. For $n \leq 19$ the simplex is optimal; the first strict product improvement occurs at $n = 20$, with dimensions $(10, 10)$.

3 Geometric reduction

We prove the geometric identities used in the optimization. They are classical or explicitly proved in the source manuscript [1]; the proofs here are included to make the present argument independent of any unverified geometric assertion. For a compact convex set C , its support function is $h_C(u) = \max_{x \in C} \langle u, x \rangle$. Support functions determine compact convex sets, and Minkowski addition adds their support functions.

Lemma 3.1 (Facet formula). *Let $K \subset \mathbb{R}^n$ be a full-dimensional polytope. For each facet F , write a_F for its $(n-1)$ -volume and ν_F for its outward unit normal. Then*

$$h_{\Pi K}(u) = \frac{1}{2} \sum_F a_F |\langle \nu_F, u \rangle|, \quad \Pi K = \sum_F [-a_F \nu_F / 2, a_F \nu_F / 2].$$

For $n = 1$ a facet is an endpoint and has volume one.

Proof. Take a unit vector u and project onto $u^\perp$. The absolute Jacobian of projection from the hyperplane of F is $|\langle \nu_F, u \rangle|$. Apart from the projections of lower dimensional faces and projection-boundary points, which have measure zero, a line parallel to u through a point of the shadow enters and leaves K through exactly two facet interiors. Summing projected facet areas therefore counts the shadow twice. This proves the first identity for unit u ; homogeneity proves it for arbitrary u . The support function of $[-v/2, v/2]$ is $|\langle v, u \rangle|/2$, proving the second identity. In dimension one the two endpoints give $h_{\Pi K}(u) = |u|$, as required. $\square$

Proposition 3.2 (Products and affine maps). *Let $A \subset \mathbb{R}^r$ and $B \subset \mathbb{R}^s$ be full-dimensional polytopes, with $r, s \geq 1$. Then*

$$\Pi(A \times B) = (|B| \Pi A) \times (|A| \Pi B), \quad R_{r+s}(A \times B) = R_r(A) R_s(B).$$

For every invertible linear map U in dimension n ,

$$\Pi(UK) = |\det U| U^{-t} \Pi K, \quad R_n(UK) = R_n(K).$$

Translations also preserve R_n .

Proof. A facet of $A \times B$ is either $F \times B$, with F a facet of A , or $A \times G$, with G a facet of B . Indeed, an exposed face is a product of exposed faces, and codimensions add. Their area-normal vectors are respectively $(|B|a_F\nu_F, 0)$ and $(0, |A|a_G\nu_G)$. The facet formula gives

$$h_{\Pi(A \times B)}(u, v) = |B|h_{\Pi A}(u) + |A|h_{\Pi B}(v),$$

the support function of the stated Cartesian product. Taking its volume and dividing by $(|A||B|)^{r+s-1}$ proves multiplicativity.

Under U , a facet's area-normal vector $a_F\nu_F$ becomes $|\det U|U^{-t}(a_F\nu_F)$. One way to see the area factor is to map a prism of unit height over F : its volume multiplies by $|\det U|$, while its height over the image facet becomes $1/\|U^{-t}\nu_F\|$. Applying the facet formula to the transformed segments proves the covariance identity. The determinant of the linear map $|\det U|U^{-t}$ has absolute value $|\det U|^{n-1}$, exactly the scale factor of $|UK|^{n-1}$. Projection volumes are translation invariant. $\square$

Proposition 3.3 (Simplex value). *For every positive integer d , any nondegenerate d -simplex satisfies $R_d(T_d) = s_d$.*

Proof. By affine invariance it suffices to use $T_d = \text{conv}(0, e_1, \dots, e_d)$, of volume $1/d!$. For $d \geq 2$ its coordinate facets have area-normal vectors $-e_i/(d-1)!$, and the opposite facet has vector $w/(d-1)!$, where $w = e_1 + \dots + e_d$. The latter identity follows either by its height and volume or by projecting that facet onto a coordinate hyperplane. The facet formula shows that ΠT_d is a translate of

$$\frac{[0, 1]^d + [0, w]}{(d-1)!}.$$

Let $Q = [0, 1]^d$. Each fiber of Q parallel to w is an interval; adding the segment $[0, w]$ extends its length by $\|w\|$ and leaves the base projection unchanged. Fubini gives

$$|Q + [0, w]| = |Q| + \|w\| |\text{proj}_{w^\perp} Q|_{d-1} = 1 + h_{\Pi Q}(w) = d + 1.$$

The last equality follows directly from the $2d$ unit-area facets of the cube. Thus

$$R_d(T_d) = \frac{(d+1)(d!)^{d-1}}{((d-1)!)^d} = \frac{(d+1)d^d}{d!} = s_d.$$

For $d = 1$, $R_1(I) = 2 = s_1$ by the stated convention. $\square$

Every simplex in a fixed dimension is an invertible affine image of the standard one. Applying its affine map independently in each factor, followed by a final ambient affine map, proves that the geometric maximization over our class is precisely the finite optimization (1.2).

4 Concavity and exact comparisons

Lemma 4.1 (Ratio monotonicity). *The sequence $f_d = \log s_d$ is concave for $d \geq 1$. Its consecutive slopes satisfy*

$$q_d := \frac{s_{d+1}}{s_d} = \frac{d+2}{d+1} \left(1 + \frac{1}{d}\right)^d, \quad q_1 = q_2 = 3 > q_3 > q_4 > \dots$$

Proof. Only monotonicity needs proof. For real $x \geq 3$, let

$$h(x) = \log \frac{x+2}{x+1} + x \log \left(1 + \frac{1}{x}\right).$$

Direct differentiation gives

$$h'(x) = \log \left(1 + \frac{1}{x}\right) + \frac{1}{x+2} - \frac{2}{x+1}, \quad h''(x) = \frac{x^2 - x - 4}{x(x+1)^2(x+2)^2} > 0.$$

Since $h'(x) \rightarrow 0$ as $x \rightarrow \infty$, strict increase of h' implies $h'(x) < 0$ throughout $[3, \infty)$. Therefore q_d strictly decreases for integer $d \geq 3$. Finally $q_1 = q_2 = 3$ and $q_3 = 80/27 < 3$. $\square$

The following comparisons are all between explicit positive rational numbers:

$$Q_- = \frac{s_{13}^{12}}{s_{12}^{13}} > 1, \quad Q_+ = \frac{s_{13}^{14}}{s_{14}^{13}} > 1, \quad (4.1)$$

$$G = \frac{s_{14}^8}{s_{13}^4 s_{12}^5} > 1, \quad H = \frac{s_{13}^6 s_{12}^4}{s_{14}^9} > 1. \quad (4.2)$$

For a compact exact certificate, the ratios lie strictly between the following rational endpoints. Every displayed integer is divided by 10^6 .

Ratio	Lower numerator	Upper numerator
Q_-	1019009	1019010
Q_+	1010131	1010132
G	1001039	1001040
H	1001185	1001186
$s_{12}^2/(s_{11}s_{13})$	1002550	1002551
$s_{14}^2/(s_{13}s_{15})$	1001959	1001960
$s_{13}^2/(s_{12}s_{14})$	1002226	1002227

These statements can be checked by integer multiplication alone. For example, the four main ratios have the prime-power forms

$$Q_- = \frac{5^2 7^{13} 11^{13} 13^{131}}{2^{290} 3^{151}}, \quad Q_+ = \frac{13^{181}}{2^{165} 3^{185} 5^{157} 7^{156} 11},$$

$$G = \frac{5^{10} 7^{101} 11}{2^{10} 3^{47} 13^{61}}, \quad H = \frac{3^{34} 13^{85}}{2^{25} 5^{11} 7^{112} 11}.$$

The accompanying standard-library Python verifier stores each numerator and denominator and both positive cross-multiplication differences. No floating-point decision occurs in these certificates.

Lemma 4.2 (The unique rate maximum). *The sequence f_d/d strictly increases up to $d = 13$ and strictly decreases thereafter. Its piecewise-linear interpolation in the numerator has the same property: if $f(x)$ is linear on each $[d, d+1]$ and equals f_d at integers, then $f(x)/x$ strictly increases on $[1, 13]$ and strictly decreases on $[13, \infty)$.*

Proof. Set $a_d = d \log q_d - f_d$. Then

$$\frac{f_{d+1}}{d+1} - \frac{f_d}{d} = \frac{a_d}{d(d+1)}, \quad a_{d+1} - a_d = (d+1)(\log q_{d+1} - \log q_d) \leq 0.$$

The two exact comparisons (4.1) say precisely that $a_{12} > 0$ and $a_{13} < 0$. Monotonicity therefore gives the asserted signs in all dimensions. On $(d, d+1)$, direct differentiation gives $(f(x)/x)' = a_d/x^2$, proving the interpolated statement as well. $\square$

Proof of Theorem 2.2. For fixed k , concavity gives

$$\sum_{i=1}^k f_{d_i} \leq k f(n/k),$$

and the balanced dimensions a and $a+1$ attain equality. Thus the maximum at that factor count is $B(n, k)$. Since $k f(n/k) = n[f(n/k)/(n/k)]$, Lemma 4.2 shows that this expression increases with k up to the real value $n/13$ and decreases thereafter. Only its two adjacent admissible integers can maximize it. For $n < 13$, the expression decreases throughout the admissible factor counts, so $k = 1$ is optimal. For $13 \leq n \leq 19$, the remaining comparisons are the seven explicit rational comparisons between one simplex and two balanced simplices. They give strict preference to one simplex; at $n = 20$ the two ten-dimensional factors win, with the exact ratio established in [1]. For completeness, the table below gives rational enclosures for the two-factor value divided by the simplex value. All endpoints are divided by 10^6 ; the lower inequality is weak and the upper inequality is strict.

n	Factor dimensions	Lower numerator	Upper numerator
13	(6, 7)	870778	870779
14	(7, 7)	893750	893751
15	(7, 8)	913688	913689
16	(8, 8)	935695	935696
17	(8, 9)	955049	955050
18	(9, 9)	976160	976161
19	(9, 10)	994936	994937
20	(10, 10)	1015230	1015231

The verifier supplies the exact numerator and denominator for each entry. At $n = 14$ the lower endpoint is exact: the ratio is $143/160$. $\square$

Corollary 4.3 (Prescribed factor and facet counts). *Among simplex products of total dimension n with exactly k factors, the maximum is $B(n, k)$. Such a product has exactly $n+k$ facets. Consequently, for each F with $n+1 \leq F \leq 2n$, the maximum among these products with exactly F facets is $B(n, F-n)$.*

Proof. The fixed-factor claim is the concavity argument in the preceding proof. A d_i -simplex has d_i+1 facets, and the facet classification in Proposition 3.2 shows that the counts add under Cartesian products. Their sum is $\sum_i (d_i + 1) = n + k$. $\square$

5 The residue argument

For $d \geq 1$, define the nonnegative rate deficit

$$\delta_d = dL - f_d.$$

In particular $\delta_{13} = 0$, $\delta_{12} = \alpha$, and $\delta_{14} = \beta$.

Lemma 5.1 (Linear lower bounds for the deficits). *For positive integer d ,*

$$\delta_d \geq \begin{cases} (13-d)\alpha, & d \leq 12, \\ 0, & d = 13, \\ (d-13)\beta, & d \geq 14. \end{cases}$$

Equality is possible only at $d \in \{12, 13, 14\}$.

Proof. For $d \leq 12$, decreasing consecutive slopes give

$$f_{13} - f_d = \sum_{j=d}^{12} \log q_j \geq (13-d) \log q_{12}.$$

Since $f_{13} = 13L$ and $\log q_{12} - L = \alpha$, this is the first bound. It is strict if $d \leq 11$, because the sum then includes a slope strictly greater than $\log q_{12}$. The argument for $d \geq 14$ uses $f_d - f_{13} \leq (d-13) \log q_{13}$ and $L - \log q_{13} = \beta$, with strict inequality at $d \geq 15$. $\square$

For any factorization of n , put

$$A = \sum_i (d_i - 13)_+, \quad B = \sum_i (13 - d_i)_+.$$

If there are k factors, then $n = 13k + A - B$. Thus $A - B \equiv r \pmod{13}$. Lemma 5.1 yields

$$nL - \sum_i f_{d_i} \geq A\beta + B\alpha \geq D_r := \min\{r\beta, (13-r)\alpha\}, \quad (5.1)$$

where $D_0 = 0$. To prove the second inequality, remove $\min(A, B)$ from both A and B , which preserves the residue and weakly lowers the positive cost. The remaining nonzero integer is either congruent to r or to $-r$ modulo 13. Its least possible positive value is therefore r or $13 - r$, respectively; an extra multiple of 13 only increases the cost.

The exact comparisons (4.2) are equivalent to

$$8\beta < 5\alpha, \quad 4\alpha < 9\beta.$$

They place the unique crossing between $r = 8$ and $r = 9$. Hence

$$D_r = \begin{cases} r\beta, & 0 \leq r \leq 8, \\ (13-r)\alpha, & 9 \leq r \leq 12. \end{cases} \quad (5.2)$$

Proof of Theorem 2.1. The rate assertion follows from Lemma 4.2. Equation (5.1) bounds every product by $\rho^n e^{-D_r}$. If $r \leq 8$ and $n \geq 14r$, the product of r fourteen-dimensional simplices and $(n - 14r)/13$ thirteen-dimensional simplices attains that bound. If $r \geq 9$ and $n \geq 12(13 - r)$, the product with $13 - r$ twelve-dimensional factors and $(n - 12(13 - r))/13$ thirteen-dimensional factors attains it. This proves the claimed formulas and their individual thresholds.

All equality conditions are strict away from the displayed choices. The first inequality in (5.1) forces every dimension to belong to $\{12, 13, 14\}$. Cancellation is strict when $A, B > 0$, so twelve- and fourteen-dimensional factors cannot coexist. Additional multiples of 13 in A or B are strictly more expensive. Finally the two residue branches never tie. This proves uniqueness of the dimension multiset.

For $1 \leq r \leq 8$, the last nonnegative dimension in that residue class which fails $n \geq 14r$ is $14r - 13$; its largest value is 99, at $r = 8$. The residue $r = 0$ has no excluded positive dimension. For $r \geq 9$, the largest excluded dimension is at most 35. Thus every $n \geq 100$ satisfies the relevant threshold. At $n = 99$, equality in the bound would require eight fourteen-dimensional factors, whose total dimension already exceeds 99. Strictness of the equality conditions proves failure of the periodic formula there, so the uniform threshold cannot be lowered. $\square$

6 Isolation and stability

Theorem 6.1 (Sharp isolation of the maximizing products). *Let $n \geq 100$. If a product of simplices K of total dimension n has a dimension multiset different from the maximizing multiset in Theorem 2.1, then*

$$R_n(K) \leq \frac{P_n}{G}, \quad G = \frac{s_{14}^8}{s_{13}^4 s_{12}^5} \simeq 1.00103981380358.$$

No larger dimension-independent factor can replace G . For every $n \equiv 8 \pmod{13}$ with $n \geq 112$, equality is attained by

$$T_{12}^5 \times T_{13}^{(n-60)/13}.$$

Proof. Write $g = \log G = 5\alpha - 8\beta$ and $h = \log H = 9\beta - 4\alpha$. The exact certificate table gives

$$\min \left\{ h, \log \frac{s_{12}^2}{s_{11}s_{13}}, \log \frac{s_{14}^2}{s_{13}s_{15}}, \alpha + \beta, 13\alpha, 13\beta \right\} > g > 0. \quad (6.1)$$

For a dimension $d \leq 11$, the remainder beyond the first bound in Lemma 5.1 is

$$\delta_d - (13 - d)\alpha = \sum_{j=d}^{12} (\log q_j - \log q_{12}) \geq \log q_{11} - \log q_{12} = \log \frac{s_{12}^2}{s_{11}s_{13}} > g.$$

For $d \geq 15$, the analogous remainder is at least $\log q_{13} - \log q_{14} = \log(s_{14}^2/(s_{13}s_{15})) > g$. Consequently any factor outside $\{12, 13, 14\}$ already produces a loss strictly greater than g beyond the universal residue bound.

It remains to consider products using only those three dimensions. Their deficit is exactly $A\beta + B\alpha$. If both A and B are positive, cancelling one from each reduces the cost by $\alpha + \beta > g$, while the remaining cost is at least D_r . If only one is positive but exceeds the least representative of its residue, the extra cost is at least 13α or 13β , also greater than g . The only remaining nonoptimal possibility is the other residue branch. The difference between the two branch costs is

$$|r\beta - (13 - r)\alpha| = |r(\alpha + \beta) - 13\alpha|.$$

As a function of integral r , its minimum occurs at one of the two integers adjacent to the crossing, namely 8 and 9. The values there are g and h , respectively, and $h > g$. Thus every nonoptimal product loses at least g in logarithmic value. Exponentiating proves the inequality.

For $r = 8$, the optimal branch uses eight fourteen-dimensional factors. The other branch uses five twelve-dimensional factors, giving the displayed product whenever $n \geq 112$. Its logarithmic deficit exceeds the optimum by exactly $5\alpha - 8\beta = g$. This proves sharpness. $\square$

Corollary 6.2 (Bounded exceptions in near maximizers). *For $n \geq 100$ and a simplex product with factor dimensions d_i , put $\varepsilon = \log(P_n/R_n(K)) \geq 0$. Then*

$$\sum_i |d_i - 13| \leq \frac{\varepsilon + D_r}{\min\{\alpha, \beta\}}.$$

In particular, within any fixed multiplicative factor of the maximum, the number of factors of dimension different from 13 is bounded independently of the total dimension.

Proof. The left side is $A + B$ and $\varepsilon + D_r = \sum_i \delta_{d_i} \geq A\beta + B\alpha \geq \min\{\alpha, \beta\}(A + B)$. $\square$

7 Periodicity and asymptotics

The preceding theorem implies the exact recurrence

$$P_{n+13} = s_{13}P_n \quad (n \geq 100), \quad (7.1)$$

and hence a rational ordinary generating function:

$$\sum_{n \geq 1} P_n z^n = \sum_{n=1}^{99} P_n z^n + \frac{\sum_{n=100}^{112} P_n z^n}{1 - s_{13} z^{13}}.$$

Here the coefficients are rational optimal values; no counting interpretation is asserted.

For $n \geq 100$ with residue r , the excess over the simplex has the exact expression

$$\frac{P_n}{s_n} = e^{-D_r} \rho^n \frac{n!}{(n+1)n^n}. \quad (7.2)$$

Stirling's expansion therefore gives, uniformly over the thirteen residues,

$$\frac{P_n}{s_n} = e^{-D_r} \sqrt{\frac{2\pi}{n}} \left(\frac{\rho}{e}\right)^n \left(1 - \frac{11}{12n} + \frac{265}{288n^2} - \frac{47839}{51840n^3} + O(n^{-4})\right). \quad (7.3)$$

In particular the optimal exponential improvement within this entire product class is $\rho/e \simeq 1.03372825553$; the subexponential prefactor is an explicit period-13 constant times $n^{-1/2}$. Stirling's formula also gives $s_d^{1/d} \rightarrow e$. The strict decrease of these rates after dimension 13 therefore proves $\rho > e$ without relying on the displayed decimal approximation.

More generally, for each fixed positive integer J , the standard logarithmic Stirling expansion [4] yields

$$\begin{aligned} \log \frac{P_n}{s_n} &= n(\log \rho - 1) - \frac{1}{2} \log n + \frac{1}{2} \log(2\pi) - D_r - \log(1 + 1/n) \\ &\quad + \sum_{j=1}^J \frac{B_{2j}}{2j(2j-1)n^{2j-1}} + O(n^{-2J-1}). \end{aligned}$$

This determines every algebraic correction explicitly. The finite periodic term is exact, and the only asymptotic input is Stirling's classical expansion.

8 Validation and further questions

The geometric identities in (1.1), the dimension-twenty counterexample, and the existence of an exponential excess are source results. The contribution here is the optimization of the product class: the unique block size 13, the all-dimensional two-candidate formula, the exact eventual residue classes and sharp threshold, the sharp isolation factor, and the periodic asymptotic prefactor. A targeted search did not locate these specific optimization statements; that search is evidence of a plausible new refinement rather than a proof of priority. None of the arguments assumes the source collection's more ambitious claims about arbitrary convex bodies.

The exact arithmetic artifact verifies seven rational interval certificates and independently computes P_n for $1 \leq n \leq 300$ by the recurrence

$$P_0 = 1, \quad P_n = \max_{1 \leq d \leq n} s_d P_{n-d}.$$

Every dimension agrees with the two-candidate theorem, and every attainable residue formula agrees as well. These finite computations check constants and implementations; the proofs above establish the claims for all dimensions.

A separate implementation retains the two largest distinct rational values at every dimension and tracks their dimension multisets. It checks all dimensions through 260, finds a unique maximizing multiset in each, and verifies the isolation bound from dimension 100 onward. Within that range its minimum observed ratio equals G exactly, at $n = 112, 125, 138, 151, 164, 177, 190, 203, 216, 229, 242, 255$. For example, it recovers

$$\text{Opt}(99) = (12, 12, 12, 12, 12, 13, 13, 13)$$

and the runner-up at dimension 112 with dimensions $(12, 12, 12, 12, 12, 13, 13, 13, 13)$. Both programs use rational arithmetic for all decisions; decimals in their reports are only displays. The full comparison outputs and source-provenance record accompany the manuscript.

Several research directions follow directly.

1. Can one exhibit a convex body K in some dimension d with $R_d(K)^{1/d} > \rho$? Such a body would improve the best exponential rate supplied by simplex products, through the same product identity.
2. Does the supremum of $R_d(K)^{1/d}$ over all dimensions and all convex bodies occur at a finite dimension? The present theorem answers this restricted to simplex factors.
3. How does the optimum change with a prescribed vertex budget or simultaneous vertex and facet budgets? The vertex count is $\prod_i (d_i + 1)$, while Corollary 4.3 solves the facet-count constraint alone. The combined problem imposes an additional nonlinear restriction on the dimension partition.
4. Can the deficit method be extended to factors drawn from several geometrically distinct families? A unique best rate would suggest eventual periodic corrections governed by a finite optimization over dimension residues, but concavity may fail.
5. Does a geometric perturbation of an optimal product improve its normalized projection-body volume while preserving its combinatorial type? The sharp gap above concerns changes of factor dimensions, and imposes no obstruction to such perturbations.
6. What are the exact second- and third-largest product values in every sufficiently large dimension? The sharp uniform gap locates one infinite family of runner-up products, but a full

residue-by-residue classification of further values would describe the local discrete optimization landscape more precisely.

7. For centrally symmetric factor families, is there an analogous unique optimal block size and finite residue rule? The relevant factor values must first be established or bounded, and they need not satisfy the concavity exploited here.
8. Can the complete argument be formalized, including the facet formula and affine covariance, so that the arithmetic certificates and the geometric reduction are checked in one proof system? The present artifact verifies the finite arithmetic and supplies ordinary mathematical proofs of the claims in all dimensions; it does not contain a Lean formalization.

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